

Low-Complexity VSB Generation and Chirp Management for VHSP Downstream Links Supporting GPON Coexistence

Christoph Füllner⁽¹⁾, Laurens Breyne⁽²⁾, Wouter Lanneer⁽²⁾, René Bonk⁽³⁾

⁽¹⁾ Nokia Bell Labs, 70469 Stuttgart, Germany, christoph.fuellner@nokia-bell-labs.com

⁽²⁾ Nokia Bell Labs, 2018 Antwerp, Belgium

⁽³⁾ Nokia Network Infrastructure Fixed Networks, 70469 Stuttgart, Germany

Abstract We propose a novel hybrid PON concept at 100 Gb/s combining vestigial sideband (VSB) generation with an easy-to-implement transmitter chirp management to enable GPON coexistence of VHSP. We show that dispersion levels from 0 to 125 ps/nm can be covered. ©2026 The Author(s)

Introduction

The ITU-T is studying Passive Optical Network (PON) technologies beyond commercialized 50G solutions in the Very High Speed PON (VHSP) supplement, targeting capacities of at least 100 Gb/s per wavelength [1]. An attractive option is to retain the use of intensity modulation / direct detection (IM/DD) with simple on-off keying (OOK) for its large optical power budget and superior cost efficiency. Such traditional IM/DD has recently been shown to support super-rated PON downstream (DS) with net rates of 100 Gb/s if operated close to the zero-dispersion wavelength (e.g., at 1315 nm) and exploiting advanced digital signal processing (DSP) [2]. Such schemes are expected to further benefit from the advent, maturing, and cost decline of broadband IM/DD components from the data center market [3].

However, to preserve coexistence with legacy GPON VHSP DS might need to operate at wavelengths ≥ 1364 nm leading to worst-case chromatic dispersion (CD) levels of 120 ps/nm accumulated over 20 km, see Figs. 1(a) and (b), as GPON blocks the range close to zero dispersion. For double-sideband (DSB) spectra as used in traditional IM/DD, power fading limits the CD tolerance of ≥ 100 GBd OOK to a symmetric window typically covering a range less than ± 50 ps/nm [4,5] when assuming a chirp-free transmitter (Tx) and a CD-related penalty of up to 3 dB [6]. Hybrid PON techniques are now considered for advancing DD-based VHSP to CD levels of 120 ps/nm:

(i) One strategy is shifting the CD tolerance window towards positive CD levels, e.g., by chirp management or engineering [6,7] or by CD pre-compensation [8]. However, achieving high loss budgets for short as well as long PON links remains a challenge with these methods and may require sophisticated techniques such as burst-like DS reception where chirp or pre-compensation filters are dynamically adjusted for each optical network unit (ONU) connection.

(ii) Optical duobinary (ODB) [9] and single-sideband (SSB) transmission can broaden the CD tolerance window and improve the overall dispersion tolerance. Multiple ways for generating SSB

spectra have been reported [10], but they are not equally attractive for PON. Adding a laser line at the spectral edge of a carrier-suppressed DSB spectrum [11] is prohibitive and digital carrier generation effectively reduces the effective number of bits available for the digital-to-analog conversion (DAC) of the data [12]. Optical suppression of one sideband after conventional DSB generation requires either a frequency-stable laser or a tunable optical filter. In times of DSP usage, the most straightforward way of SSB generation is driving an inphase-quadrature modulator (IQM) or a dual-drive Mach-Zehnder Modulator (DD-MZM) with an OOK drive signal and its Hilbert transform [10]. Yet the implementation of the associated digital filters comes at the cost of increased Tx DSP complexity. Moreover, IQMs feature high insertion losses and achieving an appropriate sideband suppression level can be challenging for uncalibrated drive arms and for cheap MZMs with poor extinction ratio (ER).

In this paper, we explore a method that falls in between both common strategies: it combines simple and easy-to-implement MZM chirp management (i) with a vestigial sideband (VSB) generation relying on a carefully chosen delay (time skew) introduced between two DD-MZM drive arms (ii) [13]. We show that 100 Gb/s PON DS links can be closed for CD levels between 0 and 125 ps/nm when considering 3 dB CD penalty. More specifically, we demonstrate that a one-sample delay can strongly reduce the CD-related penalty. Such a delay can be easily implemented in the analog or digital domain and may open a path to low-cost hybrid PON implementations.

Proposed Transmitter Concept

The proposed concept is illustrated in Figs. 1(c) and (d). On a hardware level, a DD-MZM is driven by two voltage signals V_1 and V_2 at 100 Gb/s each. The MZM is biased at its quadrature (Q) point and modulates the intensity of an optical carrier. For chirp management, the bias circuitry allows toggling between two Q points, one with a positive (V_b+) and the other one with a negative

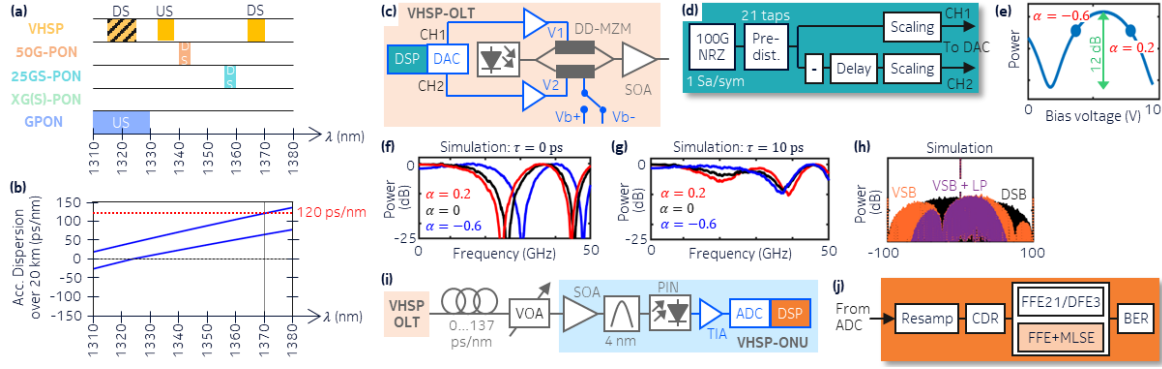


Fig. 1: (a) Potential VHSP wavelength plans. (b) Dispersion level accumulated over 20 km SSMF. (c) Proposed Tx hardware scheme. (d) Proposed Tx DSP scheme. (e) DD-MZM transfer function and chirp levels α . (f) Simulated channel response for different α and push-pull drive ($V_2 = -V_1$). (g) Simulated channel response if a delay of 10 ps is introduced between the two drive arms. (h) Simulated optical spectra for $\alpha = 0$ without (DSB) and with delay (VSB). VSB + LP refers to the case of strong Tx bandwidth limitations. (i) Receiver DSP.

(V_b) small-signal chirp parameter α , see MZM transfer function in Fig. 1(e). The MZM chirp is known to shift the position of the CD-induced fading nulls in the receive spectrum, see simulation for different α in Fig. 1(f). Note that this static and frequency-independent chirp caused by the limited extinction ratio (ER) is even present for ideal push-pull operation with complementary drive signals ($V_2 = -V_1$) generating a DSB spectrum. In our concept, the two drive signals are additionally offset in time by a digital delay τ applied before DAC. At the MZM output, this temporal mismatch constitutes a frequency-dependent phase modulation (transient and frequency-dependent chirp) besides the intensity modulation and the residual MZM chirp α . This alters the interference conditions for the two sideband-carrier mixing terms during DD and can thereby eliminate deep fades in the receive spectrum, see Fig. 1(g). A well-chosen combination of transient chirp and MZM chirp can further improve the flatness of the overall channel response. As seen from the simulated optical spectra in Fig. 1(h), the proposed technique generates a VSB spectrum, whose asymmetry becomes more pronounced by a low-pass (LP) behavior of bandlimited Tx components.

While being able to eliminate specific CD-induced spectral nulls, a transient chirp can as well cause new notches at distinct frequencies. For certain links those notches may be out-of-band as in the example in Fig. 1(g). In a back-to-back (B2B) scenario, however, a notch at $f_{\text{notch}} = 1/2\tau$ will introduce a performance penalty. Thus, since ONUs have different distances from the central office, a trade-off needs to be made between improving performance for strong CD and keeping penalties moderate for weak CD.

Experimental Setup

A proof-of-concept O-band version of the proposed Tx was set up with a 100 GS/s DAC (3-dB

BW: 25 GHz) and a 40-GHz DD-MZM. A 21-tap linear filter in the Tx DSP was used to pre-shape the bandwidth limitations of the Tx components, resulting in a three-level eye at the MZM output in case a DSB signal was generated ($\tau = 0$) [2]. Note that a small path length mismatch of the two drive arms was compensated but was not accounted for in the delays τ specified in the paper for convenience. No amplitude imbalance was introduced between the two drive arms for adjusting α [14] and their frequency-dependent phase responses were left uncalibrated. A semiconductor optical amplifier (SOA) was used to compensate the MZM loss and emitted 7 dBm of modulated power. Since the used DD-MZM was designed for C band, its static ER in O band was poor (12 dB). Moreover, $|\alpha|$ was not identical for neighboring Q points as in an ideal DD-MZM, see measured transfer function in Fig. 1(e). The MZM chirp was estimated from the overall transfer function of the Tx and a subsequent dispersive fiber with known CD profile [15].

The optical signal was launched into different fiber spools so that accumulated CD levels of up to 137 ps/nm could be assessed, see Fig. 1(h). A variable optical attenuator (VOA) allowed performing receive power sweeps. The PON receiver consisted of an SOA as pre-amplifier (gain: ~ 12 dB, noise figure: ~ 7 dB) followed by a 4-nm optical band-pass filter and a commercial 50G-class PIN-TIA. The polarization incident to the SOA was controlled to avoid minor sensitivity variations due to the polarization-dependent SOA gain. Finally, the received signal was captured using a 103 GS/s analog-to-digital converter (3-dB BW: 35 GHz). The data was then processed offline to recover the transmit sequence. The ONU DSP includes clock and data recovery, equalization and bit error ratio (BER) estimation. To overcome the strong bandwidth limitations of the overall link, we combined either a 32-tap

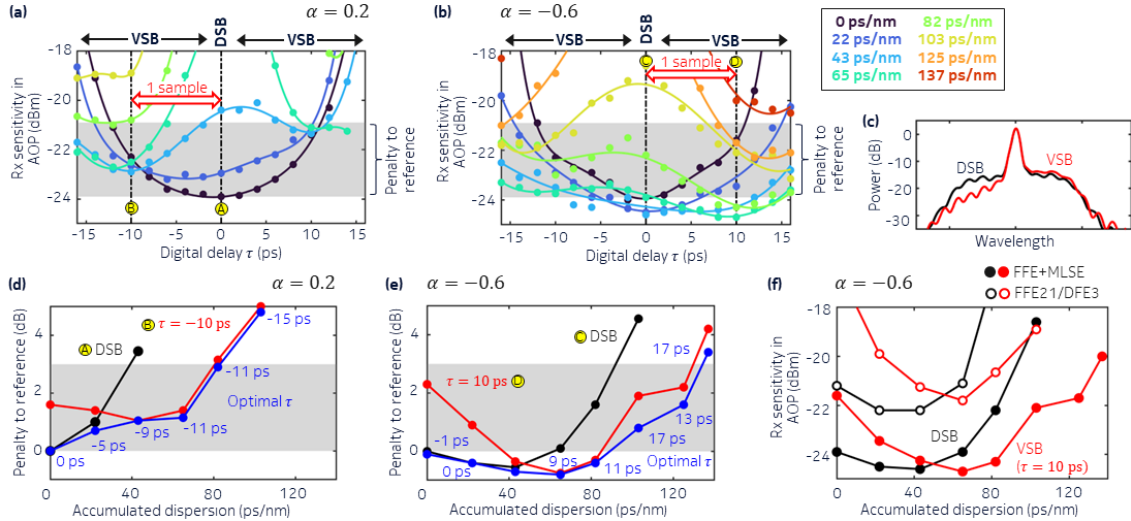


Fig. 2: Sensitivity as function of applied delay for $\alpha = 0.2$ (a) for $\alpha = -0.6$ (b). (c) Exemplary VSB spectrum. (d) CD tolerance for $\alpha = 0.2$. (e) CD tolerance for $\alpha = -0.6$. (f) Comparison of DFE and MLSE performance.

feed-forward equalizer (FFE) with a 3-tap decision-feedback equalizer (DFE), or a 4-state maximum-likelihood sequence estimation (MLSE) with a machine-learning based linear pre-equalizer structure [16].

Results and Discussion

In Figs. 2(a) and (b), the receiver sensitivity in average optical power (AOP) is displayed as a function of the delay τ introduced between the MZM drive arms. The sensitivity refers to a BER level of 2×10^{-2} . The two plots represent the two MZM Q points (Vb+ and Vb-) used, i.e. to different α . A delay of 0 ps corresponds to conventional DSB. Black markers correspond to B2B performance achieved with MLSE. All colored markers refer to a specific CD level assessed in the experiment. Polynomial fits were applied as guide for the eye.

As seen in Fig. 2(a), the B2B sensitivity degrades with increasing delay τ because spectral notches appear at low frequencies in the receive spectrum. The DSB sensitivity is -24 dBm. This B2B sensitivity level is taken as a reference for calculating sensitivity penalties. With increasing CD level, the DSB performance worsens rapidly, while a non-zero delay τ can substantially reduce the sensitivity penalty. Yet, since the undesired sideband is not fully suppressed, see exemplary optical spectrum in Fig. 2(c), the penalty grows steadily from a certain CD level onward. For a positive MZM chirp $\alpha = 0.2$, DSB transmission is limited to a positive dispersion level of <40 ps/nm when assuming 3 dB OPP. This is in line with previous electro-absorption modulated laser (EML) experiments [2]. When delaying the signal in one drive arm by one sample ('1 Sa', -10 ps) CD levels of up to ~80 ps/nm can be covered.

For a negative MZM chirp $\alpha = -0.6$ we found a similar delay dependence, yet overall better performance for strong CD levels. The best sensitivity is no longer achieved in the B2B setting

but for moderate CD levels. A penalty of 3 dB is reached between 82 and 103 ps/nm in case of DSB, which confirms that a careful choice of the used Q point can already improve the max. transmission distance. A one-sample delay (10 ps) enables closing the link for CD levels of 125 ps/nm. Figures 2(d) and (e) display the sensitivity penalties for the scenarios (A) to (D) from Figs. 2(a) and (b) as a function of the dispersion accumulated in the ODN. When comparing the penalties observed for a one-sample delay (red) with those achieved when individually optimizing τ for each CD level (blue), we found that the one-sample delay works extremely well for high CD levels and leads to tolerable penalties for weak CD. A one-sample delay could be easily implemented in the analog or digital domain without the need of a digital filter implementation. This indicates that even without a burst-like DS [8] that minimizes the path penalty per ONU, close-to-optimal performance can be achieved over a broad CD range.

Finally, we compare the performance of MLSE and DFE for DSB and VSB signals, see Fig. 2(f). We observe similarly shaped curves implying that the MLSE is mainly compensating bandwidth limitations and does only slightly affect the CD tolerance in the considered CD range.

Conclusions

We have proposed a novel hybrid PON Tx concept enabling VHSP speeds under GPON coexistence constraints. The method combines VSB generation by introducing a delay (potentially a single sample) between two DD-MZM drive arms with a simple and easy-to-implement MZM chirp management. In a proof-of-concept experiment with strong bandwidth limitations, uncalibrated drive arms, and a suboptimal DD-MZM design, we have demonstrated sensitivity penalties of less than 3 dB for dispersion levels between 0 ps/nm and 125 ps/nm.

Acknowledgements

This work was in part supported by the Flemish Government funding agency VLAIO, through the FALCON project (HBC.2024.0499).

References

- [1] ITU-T G Suppl. 88 (ex G Suppl. VHS), "Point to multipoint passive optical access system requirements and transmission technologies above 50 Gb/s per wavelength," 2025.
- [2] M. Verplaetse, C. Füllner, W. Lanneer, R. Van Rompaey, R. Bonk, and Y. Lefevre, "Super-Rated IM/DD PON Downstream Demonstration at 100G Net Rate Using Line Rates up to 124 Gb/s," *2025 European Conference on Optical Communications (ECOC)*, Copenhagen, Denmark, 2025, paper W.01.07.1, DOI: [10.1109/ECOC66593.2025.11263040](https://doi.org/10.1109/ECOC66593.2025.11263040)
- [3] V. Houtsma and D. van Veen, "Higher-Speed PONs Based On Data Center Technology and Optics," *Journal of Optical Communications and Networking*, vol. 16, no. 2, pp. A98-A104, 2024, DOI: [10.1364/JOCN.501410](https://doi.org/10.1364/JOCN.501410)
- [4] R. Bonk, E. Harstead, and A. Bender, "100 Gbit/s IM-DD Passive Optical Networks – Candidate Transceiver Technology, Parameters and Chromatic Dispersion Analysis," *2024 IEEE Opto-Electronics and Communications Conference (OECC)*, Melbourne, Australia, 2024, pp. 1-3, DOI: [10.1109/OECC54135.2024.10975423](https://doi.org/10.1109/OECC54135.2024.10975423)
- [5] V. Houtsma, W. Lanneer, K. Vijayan, R. Borkowski, and D. van Veen, "100-120G IM-DD PONs with 32 dB Power Budget and TDEC with DFE Based Reference Receiver to Ensure Interoperability," *2025 European Conference on Optical Communications (ECOC)*, Copenhagen, Denmark, 2025, paper W.01.07.2, DOI: [10.1109/ECOC66593.2025.11263313](https://doi.org/10.1109/ECOC66593.2025.11263313)
- [6] ITU-T G.9804.3, "50-Gigabit-capable passive optical networks (50G-PON): Physical media dependent (PMD) layer specification," 2021.
- [7] D. Zhang, D. Liu, and D. Nesses, "Progress of Very High Speed PON in ITU-T," *Journal of Optical Communications and Networking*, vol. 18, no. 2, pp. A112-A122, 2026, DOI: [10.1364/JOCN.573747](https://doi.org/10.1364/JOCN.573747)
- [8] G. Rizzelli, H. Kharbich, and R. Gaudino, "Multi-User Chromatic Dispersion DSP-Based Precompensation and DD Receiver for Very High Speed PON," *2025 European Conference on Optical Communications (ECOC)*, Copenhagen, Denmark, 2025, paper M.02.07.1, DOI: [10.1109/ECOC66593.2025.11262947](https://doi.org/10.1109/ECOC66593.2025.11262947)
- [9] V. Houtsma, R. Borkowski, A. Mahadevan, and D. van Veen, "Dispersion-Tolerant 100 Gb/s Downstream PON With 38 dB Power Budget and Low-Complex Equalization," *Journal of Lightwave Technology*, vol. 42, no. 11, pp. 4096-4102, 2024, DOI: [10.1109/JLT.2024.3370500](https://doi.org/10.1109/JLT.2024.3370500)
- [10] T. Bo, H. Kim, Z. Tan, and Y. Dong, "Optical Single-Sideband Transmitters," *Journal of Lightwave Technology*, vol. 41, no. 4, pp. 1163-1174, 2023, DOI: [10.1109/JLT.2022.3212473](https://doi.org/10.1109/JLT.2022.3212473)
- [11] C. Füllner, M. M. H. Adib, S. Wolf, J. N. Kemar, W. Freude, C. Koos, S. Randel, "Complexity Analysis of the Kramers-Kronig Receiver," *Journal of Lightwave Technology*, vol. 37, no. 17, pp. 4295-4307, 2019, DOI: [10.1109/JLT.2019.2923249](https://doi.org/10.1109/JLT.2019.2923249)
- [12] S. T. Le, K. Schuh, M. Chagnon, F. Buchali, R. Dischler, V. Aref, H. Buelow, and K. M. Engenhardt, "1.72-Tb/s Virtual-Carrier-Assisted Direct-Detection Transmission Over 200 km," *Journal of Lightwave Technology*, vol. 36, no. 6, pp. 1347-1353, 2017, DOI: [10.1109/JLT.2017.2779331](https://doi.org/10.1109/JLT.2017.2779331)
- [13] Y. Zhu, X. Miao, Q. Wu, L. Yin, and W. Hu, "Imbalanced Mach-Zehnder Modulator for Fading Suppression in Dispersion-Uncompensated Direct Detection System," *Electronics*, vol. 10, no. 22, pp. 1-17, 2021, DOI: [10.3390/electronics10222866](https://doi.org/10.3390/electronics10222866)
- [14] H. Kim and A. H. Gnauck, "Chirp Characteristics of Dual-Drive Mach-Zehnder Modulator With a Finite DC Extinction Ratio," *IEEE Photonics Technology Letters*, vol. 14, no. 3, 2002, pp. 298-300, DOI: [10.1109/68.986792](https://doi.org/10.1109/68.986792)
- [15] F. Devaux, Y. Sorel, and J. F. Kerdiles, "Simple Measurement of Fiber Dispersion and of Chirp Parameter of Intensity Modulated Light Emitter," *Journal of Lightwave Technology*, vol. 11, no. 12, pp. 1937-1940, 1993, DOI: [10.1109/50.257953](https://doi.org/10.1109/50.257953)
- [16] W. Lanneer and Y. Lefevre, "Machine learning-based pre-equalizers for maximum likelihood sequence estimation in high-speed PONs," *31st European Signal Processing Conference (EUSIPCO)*, Helsinki, Finland, 2023, pp. 1425-1429, DOI: [10.23919/EUSIPCO58844.2023.10289760](https://doi.org/10.23919/EUSIPCO58844.2023.10289760)